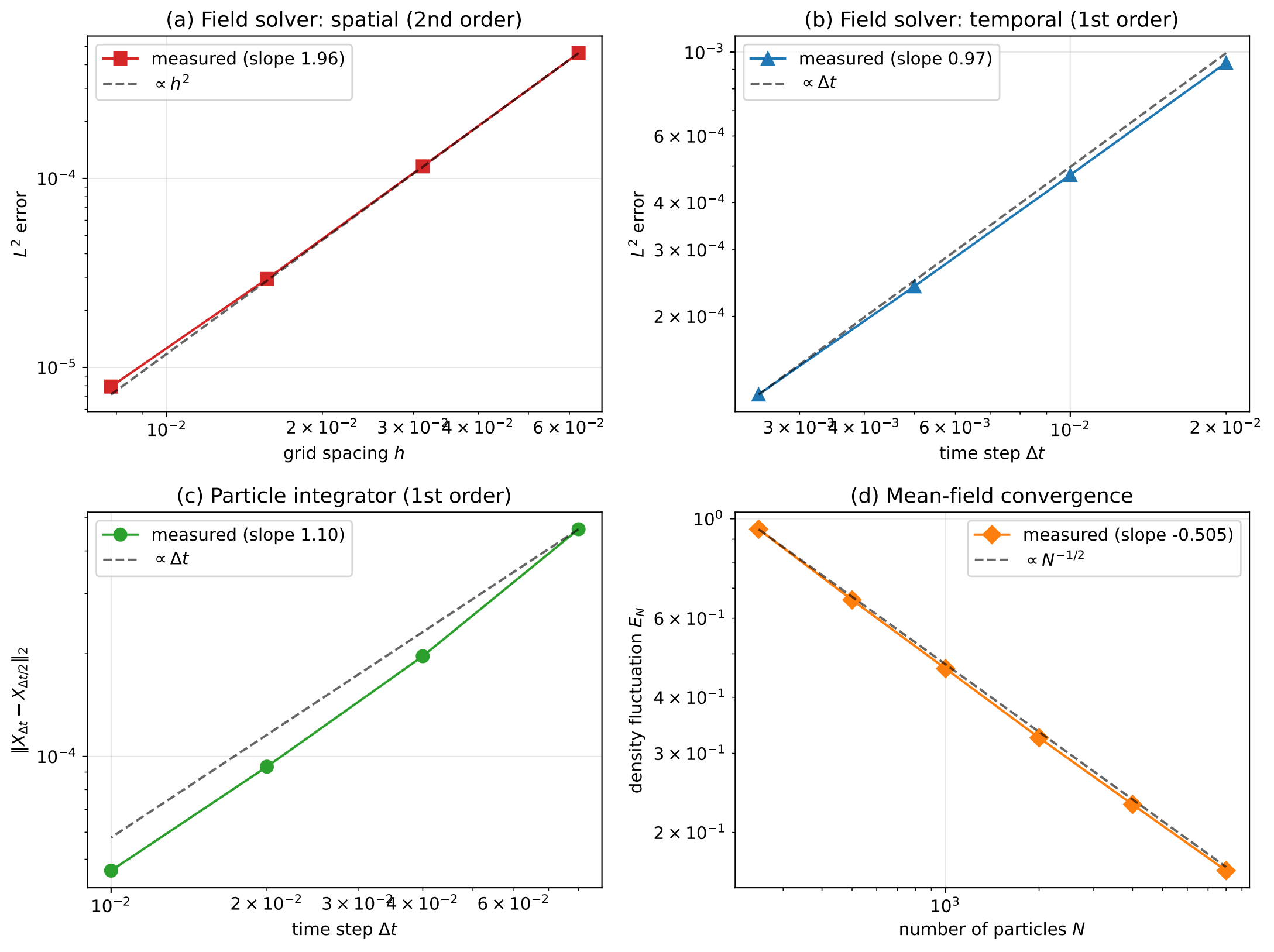
(a) Field solver: spatial (2nd order)
measured (slope 1.96)
$\propto h^2$
L^2 error
grid spacing h
(b) Field solver: temporal (1st order)
measured (slope 0.97)
$\propto \Delta t$
L^2 error
time step Δt
(c) Particle integrator (1st order)
measured (slope 1.10)
$\propto \Delta t$
$\|X_{\Delta t} - X_{\Delta t/2}\|_2$
time step Δt
(d) Mean-field convergence
measured (slope -0.505)
$\propto N^{-1/2}$
density fluctuation E_N
number of particles N